



RESEARCH PAPER

Global Ecology
and BiogeographyA Journal of
Macroecology

WILEY

The macroecology of fish migration

Dominique Alò^{1,2} | Shaw Nozaki Lacy³ | Andrea Castillo² |
Horacio A. Samaniego^{2,4,5} | Pablo A. Marquet^{1,4,5,6}

¹Departamento de Ecología, Facultad de Ciencias Biológicas, Pontificia Universidad Católica de Chile, Santiago, Chile

²Laboratorio de Ecoinformática, Instituto de Conservación Biodiversidad y Territorio, Valdivia, Chile

³Center for Climate Studies, The School for Field Studies, Puerto Natales, Chile

⁴Instituto de Sistemas Complejos de Valparaíso, Valparaíso, Chile

⁵Instituto de Ecología y Biodiversidad (IEB), Santiago, Chile

⁶Laboratorio Internacional en Cambio Global (LINCGlobal, CSIC-UC)

Correspondence

Dominique Alò, Departamento de Ecología, Facultad de Ciencias Biológicas, Pontificia Universidad Católica de Chile, Avda. Libertador Bernardo O'Higgins 340, Santiago, Chile.
Email: dominiquealo@gmail.com

Funding information

CONICYT DOCTORAL FELLOWSHIP, Grant/Award Number: 21150634

Editor: Derek Tittensor

Abstract

Aim: We still lack a consensus on the main variables driving changes in migratory strategies. Different hypotheses have been proposed: productivity, energy, environmental heterogeneity, and genetic predisposition. This work takes an integrative view and analyses migrations from a macroecological perspective estimating the extent to which different environmental variables and historic factors influence migratory life histories.

Location: Global.

Time period: Current.

Major taxa studied: Actinopterygian fishes.

Methods: Using public domain museum records, global repositories, and global measures of temperature, productivity, precipitation and heterogeneity, we spatially analysed the distribution of anadromous, catadromous, amphidromous, potamodromous and oceanodromous migratory fish using 1,676 species and compared it to 1,616 non-migratory fishes. After analysing the individual roles of productivity and temperature in shaping biodiversity, we conducted path analyses including several environmental variables and principal coordinates of phylogenetic structure (PCPS).

Results: The different migratory strategies are not evenly distributed around the globe and phylogeny is a relevant variable in shaping current patterns. Productivity is positively related to species richness, except for anadromy and potamodromy, for which we observed a unimodal curve. Temperature significantly drives migratory species richness (except for anadromy). The role of environmental heterogeneity, measured as temperature seasonality and annual range, is strongest for anadromous species, which helps explain their skewed distribution towards higher latitudes and why the kinetic energy hypothesis fails in explaining their richness patterns.

Main conclusions: Overall, migratory fish richness can be explained by the interaction of multiple variables, such as productivity, temperature, environmental heterogeneity, and the role of phylogeny, but these variables interact differentially in each strategy, diverging most for the anomalous anadromous fishes. Integrating these results into a global framework to better understand the evolutionary and ecological dynamics of migration will help with predicting responses to anthropogenic climate change.

KEYWORDS

heterogeneity, kinetic energy, metabolic theory of ecology, productivity, PCPS, PSRR, species richness

1 | INTRODUCTION

Movement is a fundamental behavioural characteristic of life on Earth. All animals move at some stage within their life spans and one type of movement, which has captivated the attention of scientists and nonexperts, is migration, a persistent and directional movement in which migrants are undistracted by stimuli, display distinct behaviour, and reallocate energy specifically to support movement (Dingle, 2014; Nathan et al., 2008). It is difficult to find common features to explain migrations since they take place in many forms, habitats and taxa. Integrating overlapping biological and behavioural questions across fields of research is as relevant today as it was when Tinbergen insisted on the importance of finding a comprehensive way to explain how specific traits were shaped by selection in the past, the environment in the present, and/or by their function and cause (Bateson & Laland, 2013; Tinbergen, 1963). Different perspectives on migration have been summarized by Dingle (2014) in a simplified model that accounts for the mutual relationship of migrants with their habitat, and the effects of the habitat on migrants through natural selection, which – acting on the genome – generates the migratory behaviour. On the basis of Dingle's inclusive framework, a series of fundamental hypotheses helps with understanding the relevance of different migratory strategies, such as: (a) *cost-benefit* approach, (b) habitat impact (through *productivity*, *energy*, *heterogeneity*), and (c) *genetic predisposition*. These hypotheses are part of a continuum of ecological and evolutionary explanations that can contribute to a holistic model of migration (Dingle, 1996; Dingle & Drake, 2007; Pulido, 2007).

The *cost-benefit* hypothesis addresses why migration evolved and persisted, since its costs are so high. If the benefit-to-cost ratio of moving exceeds that for staying, migration should be favoured by natural selection (Dingle, 1996; Lack, 1954), and fitness trade-offs occur if migration is selectively advantageous over resident life history (Bazazi et al., 2011; Gross, 1987; Simpson et al., 2006; Sword et al., 2005). According to the *cost-benefit* hypothesis, abiotic factors and habitat characteristics should influence migratory behaviour if they ultimately increase the fitness of a migratory species. Therefore, as proposed by the *productivity* hypothesis, migration is driven by the greater resource provisioning of certain habitats (Chalant et al., 2019; Gross et al., 1988) or triggered by resource scarcity (Bazazi et al., 2011; Buhl et al., 2006). Also, since the energy available from the environment (*energy* as temperature) should directly affect species diversity with predictable patterns (as proposed by the metabolic theory of ecology, MTE; A. P. Allen et al., 2006; Brown et al., 2004; Gillooly et al., 2005), migratory animals, with physiological adaptations and advantages conferred by their ability to displace, may show contrasting richness patterns in

relation to temperature. Migration confers both bioenergetic and evolutionary advantages to species adjusting to spatial and temporal habitat variation, and it is considered a metabolic necessity in seasonal environments (McLaren, 1963; Seibert, 1949; Southwood, 1962; West, 1960; Zimmerman, 1965). Life at lower temperatures invariably involves coping with variations in resource availability (i.e., *habitat heterogeneity*), and early research on the evolution of avian migrations suggested that migration might be a normal consequence of increased competition between and among species in seasonal environments (Cox, 1968) and affect life-history strategies (Beukhof et al., 2019; Mims et al., 2010; Pecuchet et al., 2017; Winemiller & Rose, 1992). Additionally, many of the morphological, physiological and behavioural traits correlated with the triggering of migration require some *genetic predisposition* that confers a tendency for a 'migratory syndrome' (Dingle, 2006; Pulido, 2011; van Noordwijk et al., 2006). Therefore, the evolutionary dynamics that might have shaped the current distribution and movement patterns among migratory species should be duly considered.

Migratory fishes represent an ideal group to test the ability of these hypotheses in explaining the occurrence of different migratory strategies across the world. This is because migratory fishes present diverse strategies and, since the vast majority of fishes are ectotherms, their dependence on environmental changes should be predictable and dependent on energetics. In particular, log-transformed fish species richness appears to increase linearly with increases in temperature, according to the available *kinetic energy* of the environment (A. P. Allen et al., 2002; Brown et al., 2004; Gillooly et al., 2001). Also, several of the oldest fish taxa have been shown to be more likely to undertake migrations, but there are simultaneously plenty of ancient taxa completely lacking migratory fishes, and this apparent mismatch offers opportunities to compare different phylogenetic hypotheses (Bloom & Lovejoy, 2014; Dodson et al., 2009; McDowall, 1997). Last, from a biogeographic perspective, some areas seem more likely to be inhabited by migratory species than others (McDowall, 1988), opening interesting avenues to relate the importance of environmental variation to the presence of migratory species.

Migration strategies in fishes have traditionally been described with reference to the aquatic environments in which they occur, with oceanodromy occurring entirely in oceans, potamodromy in freshwater, and diadromy between marine and freshwater. Three different types of diadromy have been described, depending on the life-history stages and zones in which each reproductive or feeding event occurs: anadromy involves birth in freshwater, migration to sea to grow and mature, and a return to freshwater to spawn. Catadromy implicates birth in estuarine areas, migration to freshwater to grow, and return to the ocean or estuarine areas to reproduce. Amphidromy also describes movements between freshwater

TABLE 1 Latent and manifest variables tested

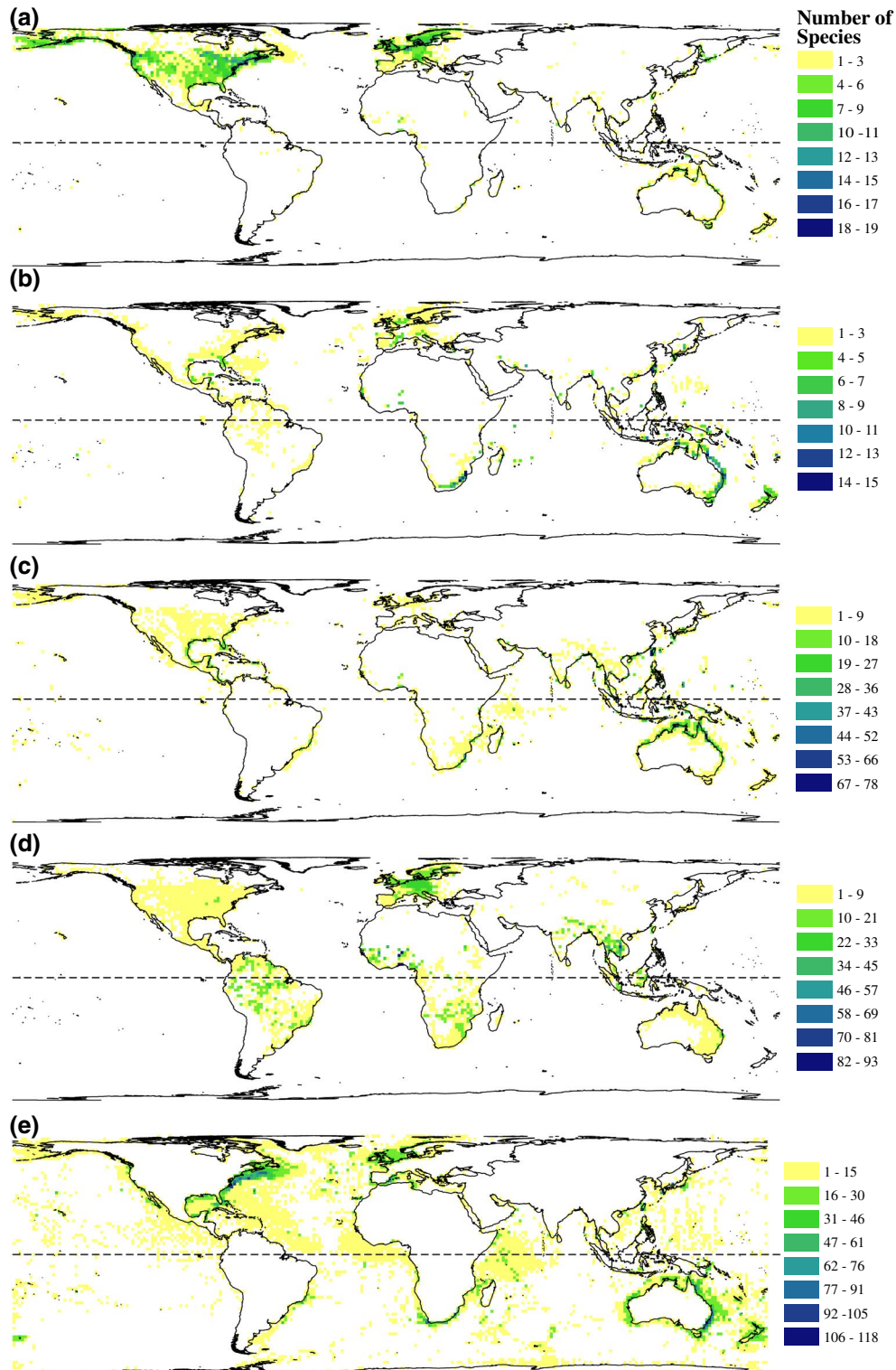
Latent variable	Manifest variable	Description	Unit	Source	Reference
Temperature	BIO1	Average annual temperature	Celsius	MERRAclim	Vega et al. (2018)
Heterogeneity	BIO2	Mean diurnal range	Celsius	MERRAclim	Vega et al. (2018)
Heterogeneity	BIO3	Isothermality	Celsius	MERRAclim	Vega et al. (2018)
Heterogeneity	BIO4	Temperature seasonality	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO5	Max temperature of warmest month	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO6	Min temperature coldest month	Celsius	MERRAclim	Vega et al. (2018)
Heterogeneity	BIO7	Temperature annual range	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO8	Mean temperature of wettest quarter	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO9	Mean temperature of driest quarter	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO10	Mean temperature of warmest quarter	Celsius	MERRAclim	Vega et al. (2018)
Temperature	BIO11	Mean temperature of coldest quarter	Celsius	MERRAclim	Vega et al. (2018)
Precipitation	BIO12	Annual precipitation	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO13	Precipitation of wettest month	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO14	Precipitation of driest month	mm	MERRAclim	Vega et al. (2018)
Heterogeneity	BIO15	Precipitation seasonality	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO16	Precipitation of wettest quarter	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO17	Precipitation of driest quarter	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO18	Precipitation of warmest quarter	mm	MERRAclim	Vega et al. (2018)
Precipitation	BIO19	Precipitation of coldest quarter	mm	MERRAclim	Vega et al. (2018)
Productivity	NPP	Average annual productivity	g C/m ² /year	MODIS	Behrenfeld and Falkowski, (1997); Zhao et al. (2005)
Phylogenetic Structure	Phylo.1	Phylogenetic filters PCPS	Eigenvector	This study	
Phylogenetic Structure	Phylo.2	Phylogenetic filters PCPS	Eigenvector	This study	
Phylogenetic Structure	Phylo.3	Phylogenetic filters PCPS	Eigenvector	This study	
Richness	SR	Species richness per grid cell	Species count	This study	

Note.: NPP = net primary productivity. Latent variable constructs represent entities that influence the dependent variable through measurable independent variables (Sanchez, 2013). Only highly correlated and homogenous variables were considered useful to be used in the path model. Only variables with Cronbach's alpha > .7, Dillon-Goldstein's rho > .7 and first eigenvalue of the indicators' correlation matrices > 1.0 were included in the models (Supporting Information Table S1, Sanchez, 2013; Vinzi et al., 2010). *SpeciesRichness* (SR) was included in the model as a direct construct (no manifest variables). The principal coordinates of phylogenetic structure (PCPS) were used as explanatory variables representing historical structure of the distribution of the migratory and resident groups. Rows in bold indicate which were the variables chosen to be included in the path models.

and saltwater, but these migrations are not driven by the purposes of feeding and/or reproduction (Dodson, 1997; McDowall, 1988). Note that, like all categorization schemes, these categories are not as discrete as the emblematic examples used to illustrate them, and

each different migratory strategy is realized in many different ways (Chapman et al., 2012).

The objective of this study is to analyse migrations from a macroecological perspective, testing the extent to which global



biodiversity patterns in migratory species can significantly be related with environmental (productivity, temperature and heterogeneity) and historical legacies. With the following scheme and using migratory fishes to test our hypotheses, first we review global migratory fish diversity to reconsider, update and complement prior work (Chalant et al., 2019; Dodson, 1997; McDowall, 1988; Nelson, 2006) using global data on species occurrences and public domain

museum records. Second, we test for a direct relationship between biodiversity patterns of migratory species using productivity and temperature. Third, we test for the existence of interactions among explanatory variables with path analyses, examining the joint relationship of several factors in shaping migratory species biodiversity, including productivity, temperature, environmental heterogeneity, and phylogenetic structure.

FIGURE 1 Global species diversity gradient richness for migratory species calculated according to occurrences downloaded from the Global Biodiversity Information Facility (GBIF) and the Ocean Biodiversity Information System (OBIS) (a) Anadromy, (b) catadromy, (c) amphidromy, (d) potamodromy, (e) oceanodromy. Hotspot ecoregions for migratory fishes (highest number of species in the region given in parentheses) classified according to the bioregionalization provided by the datasets Marine Ecoregions of the World (MEOW), Freshwater Ecoregions of the World (FEOW) and the Pelagic Provinces Of the World (PPOW) (Abell et al., 2008; Spalding et al., 2007, 2012). These were (a) Laurentian Great Lakes (20), Virginian and Gulf of Maine/Bay of Fundy (19), Eastern Bering (18); (b) Arnhem Coast to Gulf of Carpentaria (19), Natal (18), Lesser Sunda & Central and Southern Great Barrier Reef (16); (c) Southern China (75), Arnhem Coast to Gulf of Carpentaria (68), Central and Southern Great Barrier Reef (63); (d) Lower Niger-Benue and Volta (93), Gulf of Tonkin (89), Gulf of Central Guinea; (e) transitional biome of Indo-Pacific Warm Water (217), Boundary-western biome of Atlantic Warm Water (192) and transitional biome of Atlantic Warm Water (142). The line of the equator is shown as a dashed line at 0 degrees of latitude. Continental outlines extracted from ArcGIS, ESRI, Redlands, CA, USA (www.arcgis.com). Geographic coordinate system World Geodetic System (WGS) 1984 and projected to cylindrical equal area. URL: <https://www.arcgis.com/home/item.html?id=5cf4f223c4a642eb9aa7ae1216a04372>

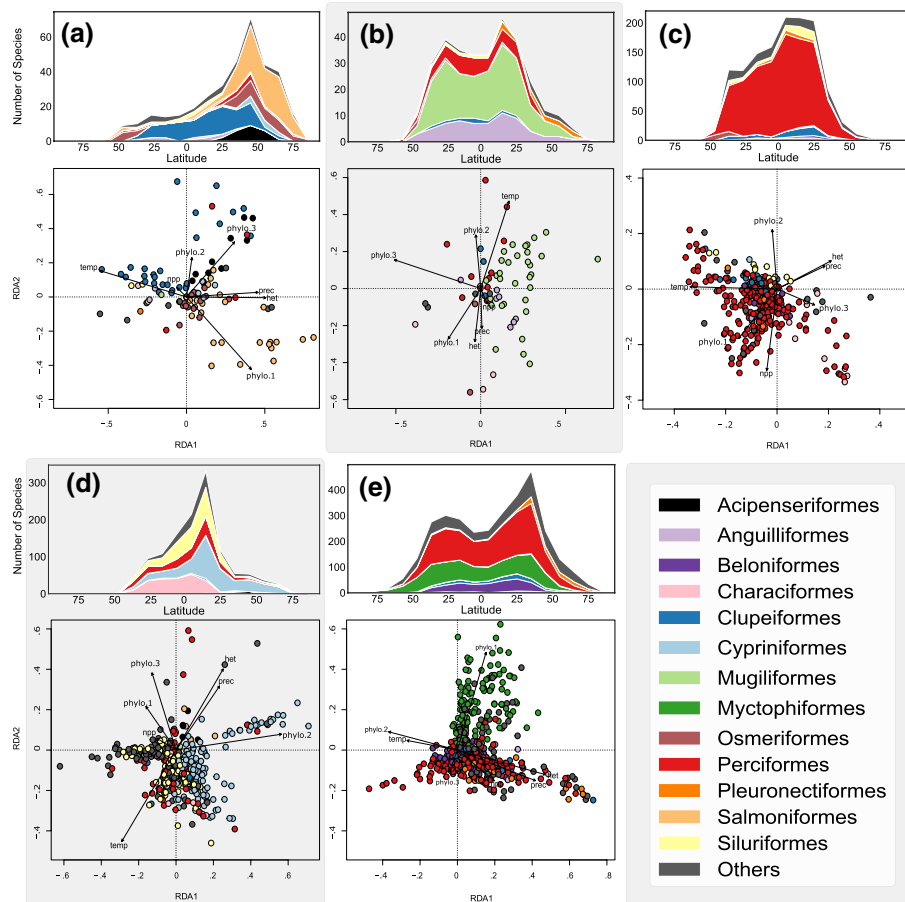


FIGURE 2 (a) Anadromy, (b) catadromy, (c) amphidromy, (d) potamodromy, (e) oceanodromy. Upper panels: species richness (SR) of each migratory group measured across segments of 10 degrees of latitude. Relative abundance of most relevant orders is shown in different colours. Lower panels: redundancy analysis (RDA) plots. These combine regression and principal components analysis (PCA; Borcard et al., 2009; Legendre & Legendre, 2012) to model selected environmental, productivity and principal coordinates of phylogenetic structure (PCPS) phylogenetic filters (shown as phylo.1, phylo.2, phylo.3). PCPS are vectors of genetic structure based on phylogenetic distance data (Pillar & Duarte, 2010) calculated from a backbone phylogenetic tree (Rabosky et al., 2018). prec = precipitation; npp = net primary productivity; temp = temperature; het = heterogeneity.

2 | METHODS

2.1 | Species data

Unless otherwise noted, all analyses were carried out using the R statistical computing language (R Core Team, 2018). Spatial analyses

and mapping employed the 'sf' and 'cartography' R libraries (Giraud & Lambert, 2016; Pebesma, 2018).

We developed a species database using multiple public information sources. First, we obtained taxonomic and migratory life-history information through Fishbase using the R library 'rfishbase' (Boettiger et al., 2012; Froese & Pauly, 2016). Species with no assigned migration/resident category were removed.

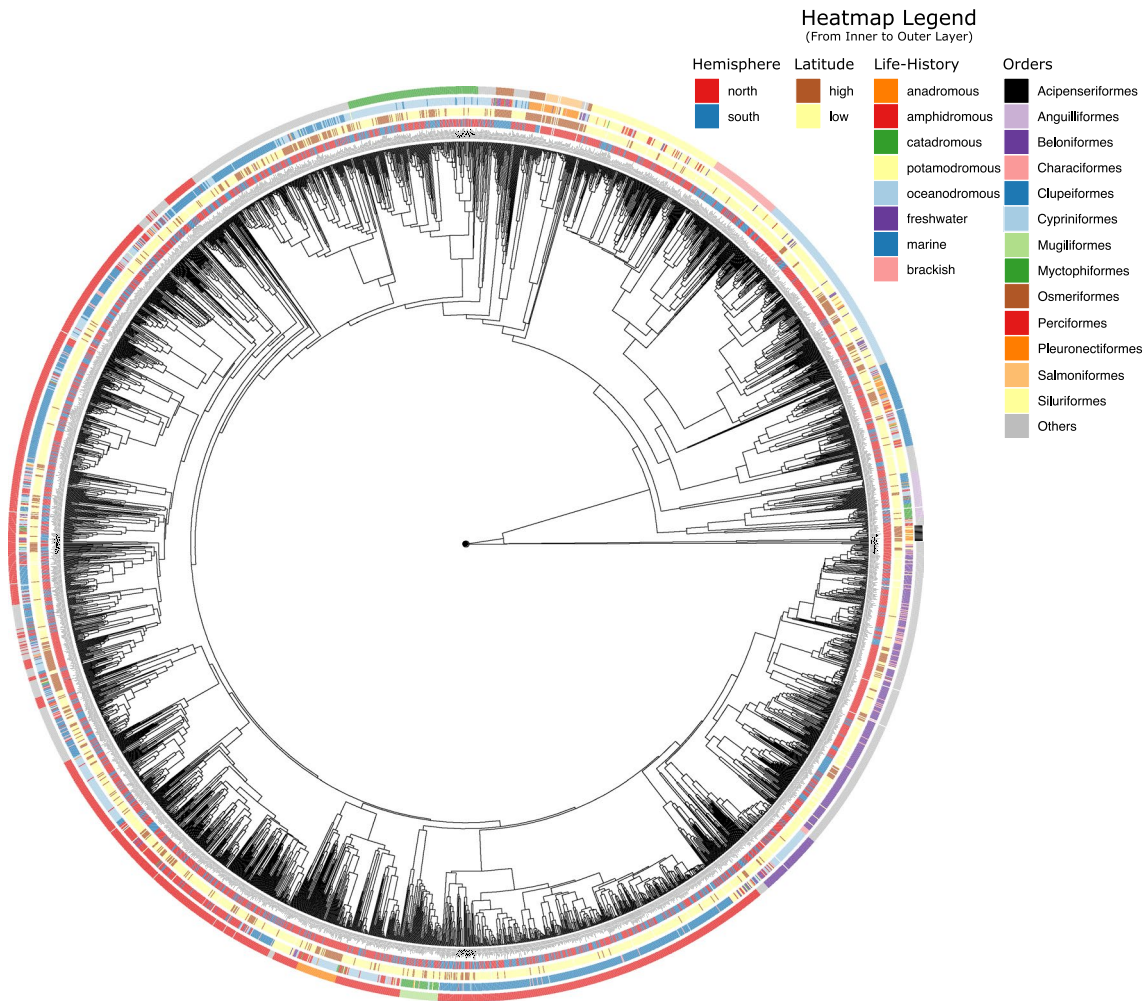


FIGURE 3 Phylogenetic tree of all Actinopterygii within the dataset. For each species, the tree shows heatmap bands corresponding to hemisphere, latitude, life history and taxonomic order. Hemisphere classification was based on whether the latitude centroid calculated for each of the species corresponded to the Southern or Northern Hemisphere. Minimum and maximum latitude for each species were used to classify them as 'low' latitude if the absolute average ≤ 30 or 'high' if absolute average > 30

Next, each species record from Fishbase was linked to its georeferenced records in the Global Biodiversity Information Facility (GBIF) and the Ocean Biodiversity Information System (OBIS), using the 'rgbif' (Chamberlain, 2017) and 'robis' (Provoost & Bosch, 2020) libraries in R, respectively. Note that, since we wanted to include phylogenetic structure as an analytical factor (see below) and there is no complete fish phylogeny, we used a subset based on the most extensive actinopterygian phylogeny available (Rabosky et al. 2018), precluding the inclusion of evolutionarily older taxa (e.g., Chondrichthyes).

Public data sources remain limited and biased by type and availability; for example, records from some areas were unavailable (e.g., northern Eurasia and Gobi Desert). Additionally, filtering is necessary to minimize errors that frequently appear when using global repositories (Garcia-Rosello et al., 2015). Therefore, we deleted all (a) records with NA values; (b) species with multiple records at the same coordinates; (c) records with identical longitude and latitude values; (d) occurrences outside the habitat range (except in coastal zones) using the shapefile of coastal and shelf

biogeographic areas from Spalding et al. (2007), which was overlapped on the freshwater (Abell et al., 2008) and pelagic (Spalding et al., 2012) ecoregion shapefiles to remove (i) oceanodromous species over continents and (ii) potamodromous over oceans, while (iii) species in desert biomes without associated water bodies (Olson et al., 2001) were deleted using a buffer zone of 50 km around the most important hydrological courses (Guan et al., 2012; Porto, 2015); (e) exotic species occurrences using the 'distribution' function in 'rfishbase' to only retain fishes in their native range using the fishing area classification shapefile (FAO, 2014); and (f) records collected before 1980. Non-migratory fishes were separated into freshwater or marine according to the regionalization provided by the freshwater ecoregions of the world (FEOW; Abell et al. 2008) and marine ecoregions of the world (MEOW; Spalding et al. 2007), respectively.

The filtered data were assigned to four different grid sizes (880, 550, 330 and 150 km²) of equal-area cells. Ultimately, the 150-km² grid, which yielded the highest model fitting values, was used to

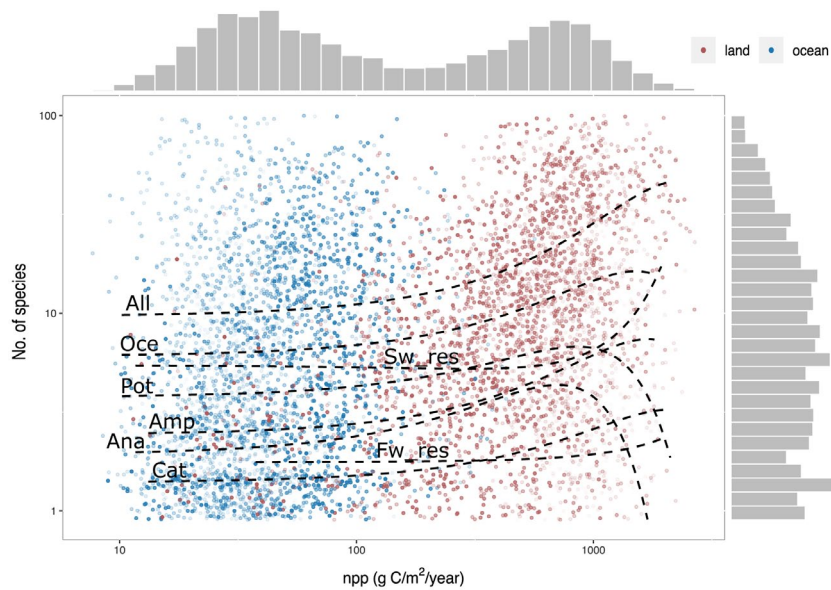


FIGURE 4 Productivity–species richness relationship (PSRR) curves show the relationship for migratory, resident and all species pooled together. Each dot represents the number of species in a particular cell of the grid and the dot colour shows whether the cell falls within ocean or land in the grid. Colour intensity is shown according to each cell's latitude, where darker colours represent higher latitudes. Some jitter was added to the points to avoid overlapping. Histograms on the margins of the figure describe the distribution of the datasets for species richness (y axis) and net primary productivity (npp; x axis). Ana = anadromy; Cat = catadromy; Amp = amphidromy; Pot = potamodromy; Oce = oceanodromy; Fw_res = freshwater residents; Sw_res = ocean residents; All = all species pooled together

generate the species richness matrices, to map each migratory strategy, and to conduct path analyses.

2.2 | Environmental data

We used Moderate Resolution Imaging Spectroradiometer (MODIS) 17A3 to reproduce global terrestrial net primary productivity (NPP; Zhao et al., 2005) and the vertically generalized production model (VGPM) of the MODIS-Aqua for estimates of oceanic NPP (Behrenfeld & Falkowski, 1997; Oregon State, 2017). More details for the NPP data are available in Supporting Information Supplemental Methods S1.

We used MERRAclim (Vega et al., 2018) to summarize 19 bioclimatic variables at 10 arc-minutes resolution, which is equivalent to 18.5 km² at the equator. Known or presumed indirect measures of productivity and heterogeneity (e.g., rainfall or temperature seasonality) were chosen as proxies to build latent variables and to test different path models.

Each environmental data layer was processed in ArcGIS, ESRI, Redlands, CA, USA, georeferenced to World Geodetic System (WGS) 1984 and reprojected to cylindrical equal area, constructing layers with annual average values.

2.3 | Phylogenetic structure

To include phylogenetic structure as an explanatory variable of migratory species richness, we used principal coordinates of

phylogenetic structure (PCPS), obtained using the method outlined by Pillar and Duarte (2010). For each migratory and resident strategy, we calculated a matrix of pairwise dissimilarities using distances averaged over the 100 full actinopterygian phylogenies of Rabosky et al. (2018) using the R package 'ape' (Paradis et al., 2004). Using the phylogenetic structure of each assemblage, we performed a principal coordinate analysis and generated PCPS orthogonal vectors that describe independent gradients in the dataset using the R package 'PCPS' (van der Windt & Duarte, 2014). For each strategy, we retained the main PCPS values that explained at least 60% of the variance, aiming at capturing deep-node phylogenetic dissimilarities. The PCPS with the highest eigenvalues typically describe larger-scale phylogenetic gradients, associated with differences at the deepest tree nodes of the phylogeny (i.e., order level, for example separating Salmoniformes from Cypriniformes). The progression to smaller PCPS eigenvalues reflects finer phylogenetic gradients related to higher nodes of the phylogenetic tree (e.g., families, genera; Duarte et al., 2012; van der Windt & Duarte, 2014). Redundancy analysis (Borcard et al., 2009), using the 'vegan' package (Oksanen et al., 2017), assessed the relationship of phylogenetic dissimilarities with the chosen PCPS values and weighed the differential importance of each environmental variable across the different clades.

2.4 | Testing the productivity hypothesis

According to the *productivity hypothesis*, NPP is fundamental in driving migrations (Chalant et al., 2019; Gross et al., 1988); however,

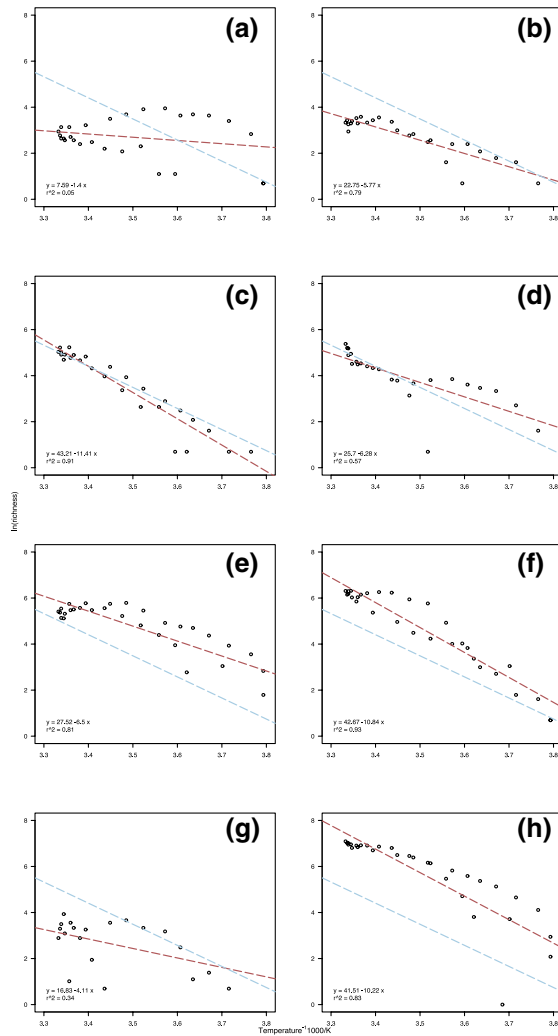


FIGURE 5 Temperature and biodiversity relationship for migratory and non-migratory species. (a) Anadromy, (b) catadromy, (c) amphidromy, (d) potamodromy, (e) oceanodromy, (f) ocean non-migratory, (g) freshwater non-migratory, (h) all species pooled together. Red dashed lines reflect data for each strategy as calculated in this study and blue dashed lines are the predicted trends according to A. P. Allen et al. (2002)

the shape of the productivity–species richness relationship (PSRR) is controversial and possibly scale-dependent, with some authors describing a unimodal (hump-shaped) relationship and others arguing that positive relationships are more common, particularly among animals (Cusens et al., 2012; Mittelbach et al., 2001). We compared species richness values in each migrating/resident category to NPP calculating the shape of each PSRR fitting a generalized linear model (GLM, *glm* function in R) with the assumption of Poisson error distribution and logarithmic link function using the equation: $\text{species richness} = a + b(\text{productivity}) + c(\text{productivity})^2$. If the quadratic component was significant ($p < .05$), we tested the significance of the hump (or pit) and its value with the null hypothesis that the quadratic coefficient is at a maximum (or minimum) using the Mitchell–Olds and Shaw test (*MOSTest* function in the ‘vegan’ package) (Leibold, 1999; Mittelbach et al., 2001). The shape of the PSRR indicates whether, and to what extent, species richness in each migratory strategy increases with habitat productivity. Several outcomes are possible: (a) a unimodal curve, (b) a u-shaped curve, (c) a monotonic positive relationship, (d) monotonic negative relationship, and (e) a neutral relationship (Chen et al., 2018; Cusens et al., 2012; Shipley, 2016).

2.5 | Testing the kinetic energy hypothesis

A. P. Allen et al. (2002) posited that the relationship between species richness and temperature should be linear along latitudinal gradients, based on the role of temperature on metabolic rate, such that the natural log of species richness should relate linearly with the inverse absolute temperature ($1,000/T_{\text{env}}$), and the slope should approximate $-E/1,000k \approx -9.0K$. To test whether migratory and non-migratory species follow this predicted trend, we used the log-transformed number of species in 5-degree latitudinal bands (36 in total) for each migration strategy and plotted them against $1,000/\text{average absolute temperature}$. The 95% confidence intervals were calculated for the slope of each linear relationship.

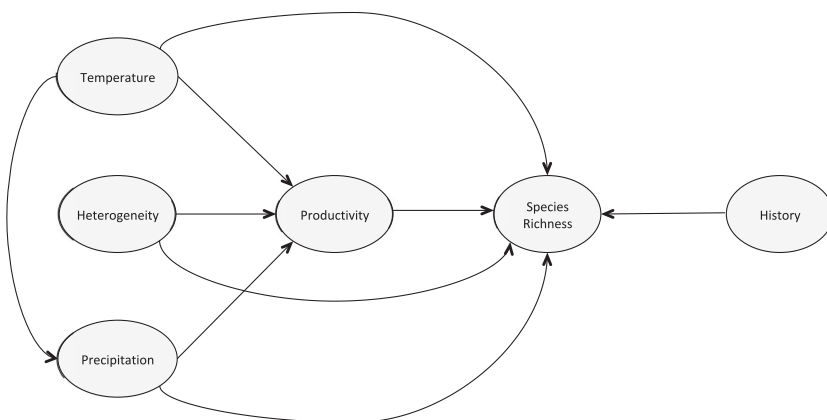


FIGURE 6 Causal diagram indicating the hypothesized relationship tested between latent variables

2.6 | Testing multiple hypotheses through path analysis

We used path analysis to define the relative importance of a priori hypotheses in shaping migratory species diversity. Trying to include the implications of several hypotheses (*productivity, kinetic energy, heterogeneity, genetic predisposition*), we constructed a model representing the simultaneous interactions of several variables influencing species richness and separately tested each dataset of migratory/resident strategies. We hypothesized that direct and indirect productivity measures (i.e., NPP and precipitation), temperature, environmental heterogeneity and phylogenetic structure interact at different levels to influence species richness in migratory species. We argue that the interplay of different variables better explains species diversity (Sanchez, 2013; Shipley, 2016) and deconstructing

species richness patterns according to migratory/resident categories can elucidate underlying processes (Marquet et al., 2004).

We hypothesized causal relationships between the factors that affect species diversity and represented these associations with a causal diagram (Shipley, 2016). A partial least-squares path modelling (pls-PM) approach was employed to test our model to accommodate non-metric data. pls-PM is a statistical, exploratory method for studying complex multivariate relationships among observed (manifest) and latent variables. All latent variables included in the model are listed in Table 1, along with their respective manifest variables. The path coefficient estimates of the hypothesized model were carried out using the 'plspm' package (Sanchez, 2013). The relevance of each assessed relationship was measured using bootstrap validation (999 samples, Supporting Information Table S1).

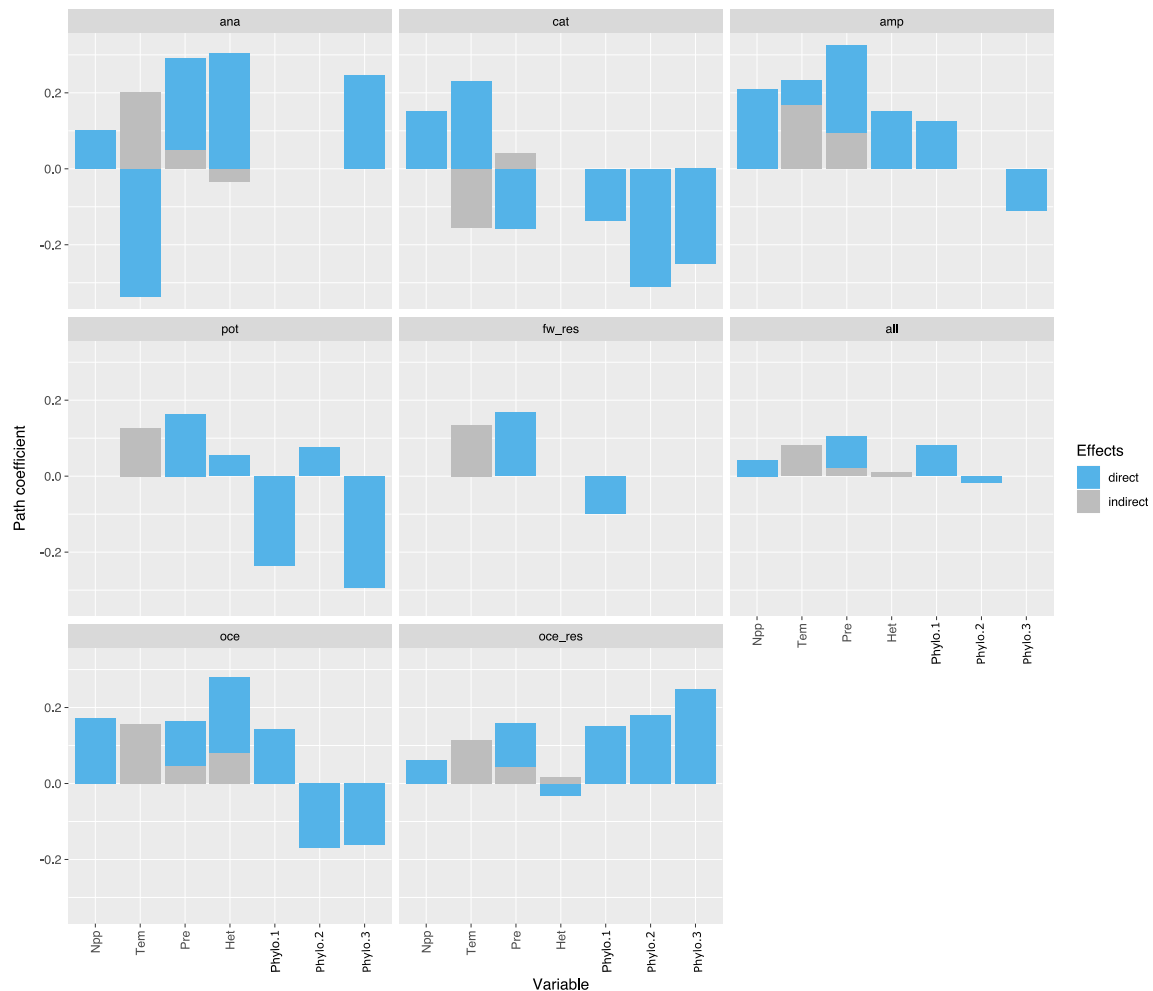


FIGURE 7 Bar plots of path coefficients for migratory and resident strategies. They represent the total effects of each latent variable as a function of direct and indirect effects. ana = anadromy; cat = catadromy; amp = amphidromy; pot = potamodromy; oce = oceanodromy; fw_res = freshwater non-migratory; oce_res = ocean non-migratory; all = all species pooled together; Npp = net primary productivity; Tem = temperature; Pre = precipitation; Het = heterogeneity; Phylo.1–Phylo.3 = principal coordinates of phylogenetic structure (PCPS).

3 | RESULTS

3.1 | Frequency of migratory life histories in fishes

The total number of fish species resulting from the Fishbase download was 33,096. Among these (total number of species given within parentheses), 2.2% were oceanodromous (624), 1.7% potamodromous (495), 1.4% amphidromous (368), 0.5% anadromous (113) and 0.2% catadromous (76). The non-migratory classification included 1,386 marine and 230 freshwater fishes. Migratory fishes represent 6.9% of global fish diversity, while diadromous fishes represent 2.4%.

3.2 | Migratory fish diversity

Migratory fishes are not evenly distributed around the globe. Maps of species richness for each migratory strategy (Figure 1) suggest that anadromous species do not increase towards the equator. The cumulative numbers of unique species in the grid cells (150 km²) contained within separated segments of 10 degrees of latitude (Figure 2, upper panels) show that the highest anadromous richness occurs between 40° N and 50° N. In contrast, maximum catadromous species richness is found between 10° S and 30° S, although the shift in maximum species richness from 0° latitude is not as pronounced as among anadromous species. In general, the highest diversity for both amphidromous and potamodromous species is between 10° N and 20° N. Oceanodromous species show uniform levels of species richness between 30° S and 30° N, with higher numbers in the Northern Hemisphere. The ecoregions (Abell et al., 2008; Spalding et al., 2007) with the highest biodiversity for each migratory strategy are listed in the legend of Figure 1 and the phylogenetic tree (Figure 3) shows latitudinal patterns from an additional taxonomic perspective.

3.3 | Productivity hypothesis

Results from PSRR show that the relationship between species richness and productivity in fishes is either unimodal (anadromous and potamodromous species) or monotonically positive (the rest of the strategies analysed) (Figure 4). The quadratic component was negative in all strategies but for resident fishes. All groups showed a significant quadratic component and were subjected to the test of Mitchell-Olds and Shaw (MOSTest), which supported a hump-shaped distribution only for anadromous and potamodromous fishes. This means increasing productivity generates higher species diversity up to a certain point (hump 696.4 and 894.3 g C/m²/year for anadromous and potamodromous species, respectively). The rest of the categories showed evidence of a positive linear (accelerating positive) relationship, meaning increasing productivity leads to continuous increases in species richness (Figure 4 and Supporting Information Table S2).

3.4 | Kinetic energy hypothesis

The linear relationship between the log of species richness and 1,000/average of the absolute temperature shows that, as predicted by A. P. Allen et al. (2002), the kinetic energy of the environment significantly influences biodiversity of all migratory and non-migratory species, except for anadromous fishes. In fact, observed values for anadromous species show more variation suggesting that other variables are important in affecting species richness of fish adopting these strategies (Figure 5, Supporting Information Table S3).

3.5 | Path analyses

According to the hypothesized causal diagram of Figure 6, the processes affecting migratory fish richness occur simultaneously and are not mutually exclusive. The proposed model was tested for all relationships listed, and the coefficients obtained in the path analyses indicate the strength and direction (positive or negative effects) of a relationship between two variables. When the bootstrap confidence intervals overlap zero, the effects of the relationship between the two variables are not considered significant. Only the latent constructs shown in bold in Table 1 were used to test the model for each strategy. Additionally, relationships with bootstrap confidence intervals overlapping zero were removed from the models (see path diagrams in Supporting Information Figure S1).

Each path analysis is supported by a goodness-of-fit (GoF) value that represents a pseudo- R^2 , where values of 0 indicate no relationship and values of 1 indicate that the variability of species richness is fully explained by the proposed variables (Henseler & Sarstedt, 2013). Results from path analyses (Figure 7 and Supporting Information Figure S1) show that net effects of *productivity* are positively associated with species richness in all but potamodromous and freshwater resident species and strongest (path coefficients $\sim .2$) in amphidromy and oceanodromy. Weaker associations are observed for anadromy, catadromy and ocean resident species. *Temperature* has a net positive relationship with species richness in all strategies except anadromy. The effects of *precipitation* on species richness showed a positive relationship for most strategies (except catadromy). The highest influence of *precipitation* on species richness was observed in anadromous, amphidromous and oceanodromous species (path coefficients: $\approx .2$). *Heterogeneity* showed the highest positive effect in anadromy (path coefficient = .3) and oceanodromy (path coefficient = .27), followed by amphidromy (path coefficient = .15). While ocean resident species showed a weak but significant interaction with *heterogeneity* (path coefficient < .08), potamodromous species showed a negative relationship (path coefficient = -.3) while catadromous and freshwater residents showed no relationship.

In our explanatory model, PCPS were hypothesized to be directly related to species richness (Figure 6). Our results show that phylogenetic structure is related to migratory strategies in different ways (Figure 6 and Supporting Information Figure S1). For example, the third PCPS (Phylo.3), which represents a phylogenetic filter

that describes a more recent evolutionary history (i.e., variation at higher nodes of the phylogeny compared to Phylo.1 and Phylo.2), is positively associated with species richness at locations exclusively occupied by anadromous species. The relationship between Phylo.3 and the migratory strategies is further explained by looking at the redundancy analysis (RDA) panels of Figure 2. RDA seeks a series of linear combinations of explanatory variables that best explain the variation in the response matrix; it can be considered as an extension of multivariate regression followed by a principal components analysis (PCA). In our case, species richness was constrained with PCPS filters and selected environmental variables. This method allowed us to more clearly visualize which clades (orders) relate to phylogeny and the environment. For example, Figure 2a for anadromous fishes shows the relationship of Phylo.3 with mainly acipenseriform fishes, whereas Figure 2d shows how some potamodromous fishes are mostly influenced by historical factors (e.g., Cypriniformes) and others (e.g., Siluriformes and Characiformes) are mostly influenced by temperature and precipitation.

Overall, all factors included in the exploratory structural model affect the biodiversity of migratory fishes. These effects can change according to the different migratory or resident strategies (Figure 7). GoF was highest for oceanodromy (.6269), anadromy (.5897) and potamodromy (.5809). With the exception of amphidromy (.5184), GoF was lowest for the 'all species' model where we pooled all strategies together (see Supporting Information Figure S1). This agrees with the initial expectations, which hypothesized that deconstructing the variance into different migratory and resident categories should allow a better justification of the underlying variability.

4 | DISCUSSION

Several hypotheses thought to influence richness in migratory species were tested. Specifically, we considered the *productivity* (Chalant et al., 2019; Gross et al., 1988) and *kinetic energy of the environment* hypotheses (A. P. Allen et al., 2002) to test whether each hypothesis could individually explain biodiversity patterns of migratory species. Since the patterns of anadromous fishes were found to be quite different from those exhibited by other categories, and in order to discern hidden relationships among variables, we integrated multivariate analyses into a path modelling framework. We tested different hypotheses through a series of exploratory path analyses, evaluating the contribution of additional variables thought to influence migratory-species richness, such as precipitation, heterogeneity, and phylogenetic structure. The most important observation stemming from this work was that migratory strategies appear to be influenced by different, identifiable environmental and phylogenetic characteristics, and the case of the anadromous fishes best illustrates this conclusion. In particular: (a) richness is strongly dependent on the *productivity* of the environment (PSRR) in migratory and resident fishes alike, although for anadromy and potamodromy this relationship shows evidence of unimodality, where species richness increases to a certain threshold and then starts decreasing; (b)

temperature is an important variable in driving biodiversity, but its influence is less relevant in anadromy, as it is only indirectly related to species richness through its causal interaction with precipitation; (c) the importance of *precipitation*, which was included in path analysis as a surrogate representing productivity of the environment, is evident in all strategies with the exception of catadromous fishes; (d) the role of *heterogeneity* is strongest for anadromy, which explains the skewed distribution patterns towards higher latitudes and low fit of the kinetic energy hypothesis; (e) the timing of events linking phylogenetic structure to species richness varies across strategies. In particular, the evolutionary gradients related to anadromous fishes show a higher affinity with the peripheral branches of the phylogeny, suggesting a more recent role of history in shaping anadromous fishes' diversity.

4.1 | Overall geographic patterns in migration strategies

Migration is costly (Gross, 1987; Somveille et al., 2015; Wikelski & Tarlow, 2003), and only a minority of fishes migrate. The observed distribution patterns of migratory fishes are likely to have been driven by evolutionary dynamics and biogeography. For example, the latitudinal richness pattern has traditionally shown how most species follow a latitudinal gradient of increasing species diversity from the poles to the equator (Brown & Lomolino 1998; Gaston 2000; Rohde 1992; but see Valdovinos et al. 2003), but anadromy peaks at 40° N and 50° N and lacks presence in the tropics or Southern Hemisphere except for a handful of species in Australia and New Zealand (Gross et al., 1988; McDowall, 1988), following an inverse pattern of latitudinal species richness compared to the rest of species analysed. This can be explained by the fact that anadromy is mostly constrained to families characterized by cold-stenothermy (McDowall, 2008) associated with large values of NPP in high-latitude coastal waters.

Other migratory strategies follow the species–latitude pattern, with maximum species richness at low latitudes in the Southern Hemisphere for catadromy, and low latitudes in the Northern Hemisphere for both amphidromy and potamodromy. While catadromy favours higher temperatures and roughly overlaps geographically with amphidromy, the latter is more constrained to tropical waters. The relatively more-constrained latitudinal distribution of potamodromy could be explained by the impacts of historical events, especially glaciation (*sensu* Svenning et al. 2015), and this appears to be the case at latitudes higher than 45° N and S, where potamodromous species richness abruptly decreases from the strongly unimodal distribution found most elsewhere in its latitudinal range (Figure 2, upper panels). In both hemispheres, river basins at these high latitudes were heavily affected by the Last Glacial Maximum, and freshwater-dependent species, being unable to take advantage of marine dispersal, would have been extirpated, except (a) in areas where exposed continental shelves allowed greater river connectivity (Dias et al., 2014; Ruzzante et al., 2008), (b) in sufficiently large north–south orientated river basins with non-glaciated extents

(Marić et al., 2014), or (c) if species were able to recolonize during glacial retreat (Griffiths, 2015; Oberdorff et al., 1997).

The diversity of oceanodromous species appears higher in the broad continental shelves of the Northern Hemisphere, possibly because of its greater shelf area compared to the South (Mackenzie et al., 2011). Also, areas influenced by strong and warm oceanic currents appear to be important sites for ocean migratory species, which are likely to be affected by seasonal alterations of current patterns (e.g., monsoon), food availability and the influence of upwellings in highly productive areas (Morais & Daverat, 2016).

Species richness patterns could also be associated with the unequal distribution across hemispheres of landmasses and coastal shelf areas (Somveille et al., 2015) as well as the extent of river area, which is greatest near the equator and at high northern latitudes (G. H. Allen & Pavelsky, 2018). For instance, Southern Hemisphere rivers are narrower south of 20° S compared to the same latitude in the Northern Hemisphere, limiting the amount of comparable habitat available. Similarly, movement restrictions originating from historic isolation between Southern Hemisphere landmasses may have contributed to the observed pattern (Lacy et al., 2019; McDowall, 2002; Unmack, 2001).

4.2 | Fish migratory strategies are influenced by phylogeny and spatial legacies

Our study indicates a strong influence of phylogeny on the current distribution of migratory species, showing that fish belonging to a particular migratory strategy tend to occur in delimited regions and often partition according to their clade structure (i.e., order, Figures 2 and 3). This suggests that each migratory strategy not only falls within a specific set of adaptations and behavioural characteristics, but also that each strategy is linked to a specific time and place of origin (Dodson et al., 2009; Lévêque et al., 2008). These phylogenetic patterns are evident in all strategies. In fact, the majority of anadromous fishes are found at high latitudes of the Northern Hemisphere and are concentrated within just a few orders, such as the Salmoniformes, Osmeriformes, Acipenseriformes, and North American Clupeiformes of the genus *Alosa*. The rest of the anadromous Clupeiformes present an exception to the anadromous pattern and are found mainly at low latitudes of the Northern Hemisphere. Catadromous fishes appear mostly divided between two main groups found at low latitudes of both hemispheres; these are the Anguilliformes and the Mugiliformes. The majority of the amphidromous fishes are found at low latitudes of the Northern Hemisphere within the order Perciformes, but also common within Siluriformes, Beloniformes and Pleuronectiformes. An exception appears within the amphidromous Osmeriformes, which mostly occur at high latitudes of the Southern Hemisphere. The majority of fishes within the orders Cypriniformes and Siluriformes are potamodromous and found at low latitudes of the Northern Hemisphere. Cyprinids, such as the Eurasian daces, seem to be an exception to this case, as they are prevalent at higher northern latitudes. Phylogenetic patterns are also

evident in oceanodromous fishes, which are mostly concentrated in Myctophiformes, Perciformes, Pleuronectiformes and Beloniformes (these are mostly found at low latitudes with the exception of the family of Exocoetidae, the flying fishes). Specific phylogenetic patterns should mostly be expected in diadromous species, as they require physiological and ontogenetic adaptations to cope with a changing environment. However, a historical signal emerged even for the oceanodromous and potamodromous species, which do not depend on physiological mechanisms to migrate through osmotic barriers. In these cases, we would expect different evolutionary and ecological drivers influencing the dependence on phylogenetic structure, such as adaptations to cope with stressful environments, competition, predation, or vicariance (in the case of potamodromous fishes). Finally, in some groups within Clupeiformes (e.g., Clupeidae, Engraulidae), Anguilliformes (e.g., Congridae), Osmeriformes (e.g., Galaxidae) and Perciformes (e.g., Moronidae, Kuhlidae, Cottidae), fishes show no uniformity of strategy. This, besides reflecting limitations in the categorization of species (discussed more in detail further below), may reflect a higher evolutionary plasticity for species within these genera.

Given the general phylogenetic patterns observed in all migratory strategies, anadromous fishes show an anomalous historical legacy. In fact, besides deviating from the expectations of the kinetic theory and showing more species in cold, high latitude areas, these species show that the evolutionary links to species richness vary significantly, according to the more peripheral branches of the phylogeny. This suggests that anadromous fishes might have been more affected by recent events, such as glaciations, and/or that their diversity may have not yet reached stable levels by mechanisms possibly linked to competitive displacement, exclusion or higher sensitivity to environmental disturbances (Huston, 1979).

4.3 | Productivity hypothesis

Continental and global scale PSRRs are usually found to be positive (Cusens et al., 2012). Our analysis shows that productivity drives species richness in migratory and resident species, although for anadromy and potamodromy, this relationship holds only to a certain point, followed by a decrease in species richness as productivity increases. This unimodal relationship suggests that, when productivity exceeds peak values, only some species benefit from increasing environmental resources. Biotic interactions, such as predation, may play an important role in anadromy and potamodromy, compared to resident and other migratory strategies, which showed monotonic positive PSRRs. Although at larger scales abiotic factors are usually considered more important in predicting patterns of species richness (Cusens et al., 2012), and reduced diversity with increasing productivity may be observed where productive, nutrient rich coastal areas are associated with cold waters that upwell from the deep. These data partially support the *productivity hypothesis*, as migratory species richness tends to increase in areas of higher productivity, with the exception of anadromous and potamodromous species.

Additionally, our analysis has shown that productivity provides a significant causal relationship for many species. Anadromy can be seen as a strategy that evolved to avoid competition in unproductive environments during certain stages of the life cycle.

Additional insights on the role of productivity can be gained by looking at the relationship of precipitation with species diversity. Rainfall has been used as a proxy to represent productivity in terrestrial systems (Lieth & Whittaker, 1975), and it is also known to impact species richness in estuaries, coastal zones, shallow areas, and upwellings of marine waters through the modulation of salinity and carbon-flow processes (Primo et al., 2009). Precipitation has shown to be a relevant climatic variable except for catadromy. It is possible that areas with increased rainfall offer more opportunities in both freshwater and marine areas, such as predator safe-havens, nursery areas and increased resource availability.

Further research on evolutionary transitions between marine and freshwater habitats may help explain the observed patterns (see Bloom, 2013; Bloom & Lovejoy, 2017; Dodson, Laroche & Lecomte, 2009).

4.4 | Kinetic energy hypothesis/metabolic theory

While most migratory species show a positive relationship with temperature, anadromous species do not. Additionally, all migratory species show a linear slope that departs from MTE predictions of $-9K$, suggesting that the temperature–species richness relationship in migratory species is different, and showing, for example, that many oceanodromous species are able to dwell at lower temperatures compared to other analysed strategies (Figure 5 and Supporting Information Table S3). Also, path analysis suggests that the temperature–species richness relationship becomes less relevant when other variables are considered, meaning that temperature cannot be the sole predictor of species presence (as seen in potamodromous species). Water currents and temperature seasonality, coupled with thermoregulation, heterogeneity in nutrient availability, and specific adaptations to cope with environmental harshness (therefore phylogeny), could also be driving species richness. The strong departure from the *kinetic energy* hypothesis in some species warrants further research into the role of metabolism regulating biological activities. Even if cold-stenothermal species are excluded (i.e., salmonids), anadromous fishes still show a marked preference for cold waters (McDowall, 2008). Among oceanodromous fishes, partial endothermy occurs in tunas; for example, adult Atlantic bluefin tuna (*Thunnus thynnus*) can maintain relatively stable body temperatures (c. 25 °C) even when traversing waters as cold as 3 °C (Carey et al., 1971), which allows them to take advantage of resource-rich waters between distant ecoregions (Bayliff et al., 1991). The discovery of a completely warm-blooded fish, the oceanodromous *Lampris guttatus*, has shed more light on the key evolutionary adaptations needed to cope with cold waters in this evolutionary line (Porch, 2005; Wegner et al., 2015). Although these forms of endothermy arose independently and are driven by different physiological processes

(Dickson & Graham, 2004), further research, looking for adaptations at the interface with a warm-blooded metabolism, may help to understand the extent to which endothermy has played a role in the evolution and maintenance of certain lineages of fishes.

4.5 | Limitations and future studies

These data were obtained from public, global repositories; therefore spatial, temporal and categorization bias can affect results. For example, there was no geospatial data available for northern Eurasia (i.e., Russian Federation, where native migratory species, such as the endangered potamodromous *Hucho taimen*, are known to exist), and the distribution of fishes from China was highly imprecise (nearly all data were concentrated in the Gobi Desert and removed during data cleaning). Also, the effect of choosing a specific bioclimatic database among the different alternatives can be biased and could result in inconsistencies in model predictions in some areas. We used the recently published MERRAclim dataset, because it provides a uniform coverage of terrestrial and ocean climatic variables and it has been used in building and comparing species distribution models in both marine and terrestrial organisms (Morales-Barbero & Vega-Álvarez, 2019). Using atmospheric variables to explain the biogeography of aquatic species can be seen as an improbable assumption and restrict the use of variables with greater explanatory power (e.g., Bio-ORACLE for oceanic species; Assis et al., 2017). However, the use of surface air temperature as a proxy for water temperature has been used widely in freshwater fish ecology (Lacy et al., 2019; Oberdorff et al., 1995) and should function in a similar manner for sea-surface temperature, which correlates with surface air temperature across large ocean areas (Cayan, 1980; Jaswal et al., 2012). Additionally, given that our study focused exclusively on species for which there was a large phylogenetic dataset (i.e., actinopterygians), we left out a potentially very important portion of migratory species (e.g., chondrichthyans), raising questions about the robustness of the relationships found when considering a broader taxonomic group.

Furthermore, besides the fact that some species may not have yet been described as migratory, the categorization across multiple phylogenetic lineages into five migratory behaviours requires the implicit recognition that there are spectra within each migratory behaviour. For instance, anadromy includes *Lovettia sealii* (Tasmanian whitebait), who only migrate to and from estuary reaches of rivers, and catadromy includes *Pseudaphritis urvillii* (congolli), where migration occurs only among females, with males being marine-resident (Schmidt et al., 2013). Similarly, the ranges and cues within oceanodromy and potamodromy are also quite varied, with significant implications for migration ranges and body size (Radinger & Wolter, 2014; Woolnough et al., 2009). Furthermore, the breadth of these examples indicate that the ‘migratoriness’ of a species likely has a strong evolutionary component, given the importance of genetics and epigenetics in governing the physiological and behavioural changes associated with migration (Baerwald et al., 2016; Hecht et al., 2012). Categorizations can also overlap, clouding or challenging the

chosen scheme, with several fishes demonstrating the capacity for multiple types of migratory behaviour, and (depending on habitat) they can be classified in several categories. For example, *Galaxias maculatus* (known as puye in Chile and inanga in New Zealand) is a facultative migrant and is classified as amphidromous, catadromous, potamodromous, and non-migratory (Alò et al., 2019; Górski et al., 2018; Ruiz-Jarabo et al., 2016). Similarly, many salmonids are partial migrants having both migratory and resident forms (e.g., *Oncorhynchus mykiss*; Baerwald et al., 2016; Ferguson et al., 2019; Hecht et al., 2012), but are often defined as anadromous regardless of the presence of any connection to a marine setting (Arostegui & Quinn, 2019). Partial migration is a rather common phenomenon among fishes. Many studies are available but only a few analyse the proximate and ultimate mechanisms that promote it. Partial migrants can serve as a great lens for studying migration because experiments can test how different drivers affect the probability and propensity of migration (Chapman et al., 2012; Glebe & Leggett, 1981; Peiman et al., 2017).

Finally, evolutionary heritage is only assessed in the context of modern-day continental positions and climates (Albert & Reis, 2011). Incorporating increasingly accurate bioclimatic and species occurrence datasets, complete phylogenies with speciation and extinction rates, and palaeogeography could help uncover additional pieces of evidence that could elucidate further the factors that have led to the current distributions of species richness in migratory species.

4.6 | Concluding remarks

Fish migrations are driven by multiple ecological and evolutionary processes. We could not isolate single factors as uniquely responsible for migratory fish richness. We tested whether specific environmental characteristics could be pinpointed as responsible for explaining species richness in several migratory strategies. We found that *productivity*, *temperature* and *heterogeneity* interact to affect the richness of migratory species. We also highlighted that the role of history has a strong impact in shaping the global distribution of these species. There are different patterns for each migratory strategy, and multiple environmental variables can relate to migratory strategies in different ways. For example, the role of *heterogeneity* is much stronger in driving species richness in anadromy than the other strategies, while the role of *productivity* seems important in driving diversity of many species.

Increasing our understanding of how these different migratory strategies interact with spatial and environmental variables as well as the legacies of their evolutionary histories may help predict their responses to climate change and human disturbances. Migrating fishes represent only 6.9% of the total fish diversity, and some are severely challenged from a conservation perspective.

The consequences of ongoing global climate change, land-use change, invasive species, river alterations and water obstructions

(Arthington et al., 2016) can be expected to be particularly troubling for many migratory species, especially for anadromous species, which have a higher number of critically endangered species (8.57%) compared to the average of the rest of the strategies (1.56%) (IUCN, 2015).¹ Worldwide, many migratory fishes constitute important fisheries, such as diadromous salmonids, clupeids, sturgeons, or marine fishes, such as the Scorpanidae family, a globally important human food source with a cosmopolitan distribution and high dispersal capabilities influenced by their migratory behaviour (Gonzalez et al., 2008). Therefore, some of the relevant characteristics that make a fish species commercially important may also be relevant from the standpoint of migration and warrant further research (see data-driven approach of Nyboer & Chapman, 2013).

This work sought to test different existing hypotheses (*productivity*, *kinetic energy*, *environmental heterogeneity* and *genetic predisposition*) in support of developing a holistic model of migration. Our results insert well into a global framework, showing how current biodiversity patterns discerned among different migratory strategies can be explained by the joint effect of multiple environmental and productivity variables, and, additionally, how biodiversity is influenced by the role of historical factors constrained by phylogeny. Migration can be considered an innate animal behaviour, the expression of which can be turned on or off. This complex conditional trait mutually depends on several ecological and evolutionary causal factors. In particular, different movement strategies depend on environmental clues that, given enough time, geography, and vicariance events, act on morphological and physiological adaptations, ultimately resulting in phylogenetic patterns and biogeographic constraints. Further examining the role that each factor might have had on different taxa will improve predictions on how, to what extent, and how fast organisms will adapt to current and anthropogenically induced environmental changes (Dingle, 2006, 2014; Dingle & Drake, 2007; van Noordwijk et al., 2006; Pulido, 2007; Pulido et al., 1996; Zink, 2011).

ACKNOWLEDGMENTS

D. Alò acknowledges support from Comisión Nacional de Investigación Científica y Tecnológica (CONICYT) Doctoral Fellowship 21150634 and helpful discussions with members of her doctoral committee: Evelyn Habit, Mauricio Lima and Eduardo Palma.

DATA AVAILABILITY STATEMENT

All data supporting the results are available from the Dryad repository: <https://doi.org/10.5061/dryad.6hdr7sqz8>.

ORCID

Dominique Alò  <https://orcid.org/0000-0003-0579-4156>

Pablo A. Marquet  <https://orcid.org/0000-0001-6369-9339>

ENDNOTE

¹ Information downloaded using 'R' package, library 'rredlist' (Chamberlain, 2018)

REFERENCES

- Abell, R., Thieme, M. L., Revenga, C., Bryer, M., Kottelat, M., Bogutskaya, N., Coad, B., Mandrak, N., Balderas, S. C., Bussing, W., Stiassny, M. L. J., Skelton, P., Allen, G. R., Unmack, P., Naseka, A., Ng, R., Sindorf, N., Robertson, J., Armijo, E., ... Petry, P. (2008). Freshwater ecoregions of the world: A new map of biogeographic units for freshwater biodiversity conservation. *BioScience*, 58, 403–414.
- Albert, J. S., & Reis, R. E. (2011). *Historical biogeography of neotropical freshwater fishes*. University of California Press.
- Allen, A. P., Brown, J. H., & Gillooly, J. F. (2002). Global biodiversity, biochemical kinetics, and the energetic-equivalence rule. *Science*, 297, 1545–1548.
- Allen, A. P., Gillooly, J. F., Savage, V. M., & Brown, J. H. (2006). Kinetic effects of temperature on rates of genetic divergence and speciation. *Proceedings of the National Academy of Sciences USA*, 103, 9130–9135.
- Allen, G. H., & Pavelsky, T. M. (2018). Global extent of rivers and streams. *Science*, 588, 585–588.
- Alò, D., Correa, C., Samaniego, H. A., Krabbenhoft, C. A., & Turner, T. F. (2019). Otolith microchemistry and diadromy in Patagonian river fishes. *PeerJ*, 7, e6149.
- Arostegui, M. C., & Quinn, T. P. (2019). Reliance on lakes by salmon, trout and charr (*Oncorhynchus*, *Salmo* and *Salvelinus*): An evaluation of spawning habitats, rearing strategies and trophic polymorphisms. *Fish and Fisheries*, 20(4), 775–794.
- Arthington, A. H., Dulvy, N. K., Gladstone, W., & Winfield, I. J. (2016). Fish conservation in freshwater and marine realms: status, threats and management. *Aquatic Conservation: Marine and Freshwater Ecosystems*, 26, 838–857.
- Assis, J., Tyberghein, L., Bosh, S., Verbruggen, H., Serrao, E. A., & De Clerck, O. (2017). Bio-ORACLE v2.0: Extending marine data layers for bioclimatic modelling. *Global Ecology and Biogeography*, 27, 277–284.
- Baerwald, M. R., Meek, M. H., Stephens, M. R., Nagarajan, R. P., Goodbla, A. M., Tomalty, K. M. H., Thorgaard, G. H., May, B., & Nichols, K. M. (2016). Migration-related phenotypic divergence is associated with epigenetic modifications in rainbow trout. *Molecular Ecology*, 25, 1785–1800.
- Bateson, P., & Laland, K. N. (2013). Tinbergen's four questions: An appreciation and an update. *Trends in Ecology and Evolution*, 28, 712–718.
- Bayliff, W. H., Ishizuki, Y., & Deriso, R. B. (1991). Growth, movement, and attrition of northern bluefin tuna, *Thunnus thynnus*, in the Pacific Ocean, as determined by tagging. *Inter-American Tropical Tuna Commission Bulletin*, 20(1), 1–94.
- Bazazi, S., Romanczuk, P., Thomas, S., Schimansky-Geier, L., Hale, J. J., Miller, G. A., Sword, G. A., Simpson, S. J., & Couzin, I. D. (2011). Nutritional state and collective motion: From individuals to mass migration. *Proceedings of the Royal Society B: Biological Sciences*, 278, 356–363.
- Behrenfeld, M. J., & Falkowski, P. G. (1997). Photosynthetic rates derived from satellite-based chlorophyll concentration. *Limnology and Oceanography*, 42, 1–20.
- Beukhof, E., Frelat, R., Pecuchet, L., Maureaud, A., Dencker, T. S., Sölmundsson, J., Punzon, A., Primicerio, R., Hidalgo, M., Möllmann, C., & Lindegren, M. (2019). Marine fish traits follow fast-slow continuum across oceans. *Scientific Reports*, 9, 1–9.
- Bloom, D. D. (2013). *Transitions between marine and freshwaters in fishes: Evolutionary pattern and process*, Doctoral Dissertation. University of Toronto. https://tspace.library.utoronto.ca/bitstream/1807/35169/11/Bloom_Devin_D_201306_PhD_thesis.pdf
- Bloom, D. D., & Lovejoy, N. R. (2014). The evolutionary origins of diadromy inferred from a time-calibrated phylogeny for Clupeiformes (herring and allies). *Proceedings of the Royal Society B: Biological Sciences*, 281, 20132081.
- Bloom, D. D., & Lovejoy, N. R. (2017). On the origins of marine-derived freshwater fishes in South America. *Journal of Biogeography*, 4(9), 1927–1938.
- Boettiger, C., Lang, D. T., & Wainwright, P. C. (2012). rfishbase: exploring, manipulating and visualizing FishBase data from R. *Journal of Fish Biology*, 81, 2030–2039.
- Borcard, D., Gillet, F., & Legendre, P. (2009). *Numerical ecology with R*. New York: Springer-Verlag.
- Brown, J. H., Gillooly, J. F., Allen, A. P., Savage, V. M., & West, G. B. (2004). Toward a metabolic theory of ecology. *Ecology*, 85, 1771–1789.
- Brown, J. H., & Lomolino, M. V. (1998) *Biogeography* (2nd ed.). Sunderland, MA: Sinauer.
- Buhl, J., Sumpter, D. J. T., Couzin, I. D., Hale, J. J., Despland, E., Miller, E. R., & Simpson, S. J. (2006). From disorder to order in marching locusts. *Science*, 312, 1402–1406.
- Carey, F. G., Teal, J. M., Kanwisher, J. W., Lawson, K. D., & Beckett, J. S. (1971). Warm-bodied fish. *American Zoologist*, 11(1), 137–145.
- Cayan, D. R. (1980). Large-scale relationships between sea surface temperature and surface air temperature. *Monthly Weather Review*, 108, 1293–1301.
- Chalant, A., Jézéquel, C., Keith, P., & Hugué, B. (2019). The global geography of fish diadromy modes. *Global Ecology and Biogeography*, 28(9), 1272–1282.
- Chamberlain, S. (2017) *rgbif: Interface to the global "Biodiversity" information facility "API"*. R package version 0.9.8. <https://CRAN.R-project.org/package=rgbif>
- Chamberlain, S. (2018) *rredlist: "IUCN" red list client. R package*. <https://CRAN.R-project.org/package=rredlist>
- Chapman, B. B., Hulthén, K., Brodersen, J., Nilsson, P. A., Skov, C., Hansson, L. A., & Brönmark, C. (2012). Partial migration in fishes: Causes and consequences. *Journal of Fish Biology*, 81, 456–478.
- Chen, A., Fang, J., Oswalt, C. M., Jo, I., Guo, Q., Wardle, D. A., Fei, S., & Bockerhoff, E. G. (2018). Impacts of climate on the biodiversity-productivity relationship in natural forests. *Nature Communications*, 9(1), 1–7.
- Cox, G. W. (1968) The role of competition in the evolution of migration. *Evolution*, 180–192.
- Cusens, J., Wright, S. D., McBride, P. D., & Gillman, L. N. (2012). What is the form of the productivity-animal-species-richness relationship? A critical review and meta-analysis. *Ecology*, 93, 2241–2252.
- Dias, M. S., Oberdorff, T., Hugué, B., Leprieux, F., Jézéquel, C., Cornu, J. F., Brosse, S., Grenouillet, G., & Tedesco, P. A. (2014). Global imprint of historical connectivity on freshwater fish biodiversity. *Ecology Letters*, 17, 1130–1140.
- Dickson, K. A., & Graham, J. B. (2004). Evolution and consequences of endothermy in fishes. *Physiological and Biochemical Zoology*, 77, 998–1018.
- Dingle, H. D. (1996). *Migration: The biology of life on the move* (2nd ed.). : Oxford University Press.
- Dingle, H. D. (2006). Animal migration: Is there a common migratory syndrome? *Journal of Ornithology*, 147, 212–220.
- Dingle, H. D. (2014). *Migration: The biology of life on the move* (2nd ed.). : Oxford University Press.
- Dingle, H. D., & Drake, V. A. (2007). What is migration? *BioScience*, 57, 113–121.
- Dodson, J. J. (1997). Fish Migration: an evolutionary perspective. In J. J. Godin (Ed.), *Behavioral Ecology of Teleost Fishes* (pp. 10–36). Oxford University Press.
- Dodson, J. J., Laroche, J., & Lecomte, F. (2009). Contrasting evolutionary pathways of anadromy in euteleostean fishes. *American Fisheries Society Symposium*, 69, 63–77.
- Duarte, L. D. S., Prieto, P. V., & Pillar, V. D. (2012). Assessing spatial and environmental drivers of phylogenetic structure in Brazilian Araucaria forests. *Ecography*, 35, 952–960.
- FAO (2014). *FAO statistical areas for fishery purposes*. FAO Fisheries and Aquaculture Department [online]. <http://www.fao.org/fishery/area/search/en>
- Ferguson, A., Reed, T. E., Cross, T. F., McGinnity, P., & Prodöhl, P. A. (2019). Anadromy, potamodromy and residency in brown trout *Salmo trutta*: The role of genes and the environment. *Journal of Fish Biology*, 95, 692–718.

- Froese, R. & Pauly, D. (eds) (2016). FishBase (version Aug 2016). *Species 2000 & ITIS catalogue of life*, 31st October 2016. Naturalis.
- García-Rosello, E., Guisande, C., Manjarres-Hernández, A., González-Dacosta, J., Heine, J., Pelayo-Villamil, P., González-Vilas, L., Vari, R. P., Vaamonde, A., Granado-Lorencio, C., & Lobo, J. M. (2015). Can we derive macroecological patterns from primary global biodiversity information facility data? *Global Ecology and Biogeography*, 24, 335–347.
- Gaston, K. J. (2000). Global patterns in biodiversity. *Nature*, 405, 220–227.
- Gillooly, J. F., Allen, A. P., West, G. B., & Brown, J. H. (2005). The rate of DNA evolution: Effects of body size and temperature on the molecular clock. *Proceedings of the National Academy of Sciences USA*, 102, 140–145.
- Gillooly, J. F., Brown, J. H., West, G. B., Savage, V. M., & Eric, L. (2001). Effects of size and temperature on metabolic rate. *Science*, 293, 2248–2251.
- Giraud, T., & Lambert, N. (2016). cartography: Create and integrate maps in your R Workflow. *The Journal of Open Source Software*, 1, 54.
- Glebe, B. D., & Leggett, W. C. (1981). Latitudinal differences in energy allocation and use during the freshwater migrations of American Shad (*Alosa sapidissima*) and their life history consequences. *Canadian Journal of Fisheries and Aquatic Sciences*, 38, 806–820.
- González, E. G., Beerli, P., & Zardoya, R. (2008). Genetic structuring and migration patterns of Atlantic bigeye tuna, *Thunnus obesus* (Lowe, 1839). *BMC Evolutionary Biology*, 8, 252.
- Górski, K., Habit, E. M. M., Pingram, M. A. A., & Manosalva, A. J. J. (2018). Variation of the use of marine resources by *Galaxias maculatus* in large Chilean rivers. *Hydrobiologia*, 814, 61–73.
- Griffiths, D. (2015). Connectivity and vagility determine spatial richness gradients and diversification of freshwater fish in North America and Europe. *Biological Journal of the Linnean Society*, 116, 773–786.
- Gross, M. R. (1987). Evolution of diadromy in fishes. *American Fisheries Society Symposium*, 1, 14–25.
- Gross, M. R., Coleman, R. M., & McDowall, R. M. (1988). Aquatic productivity and the evolution of diadromous fish migration. *Science*, 239, 1291–1293.
- Guan, W. W., Bol, P. K., Lewis, B. G., Bertrand, M., Berman, M. L., & Blossom, J. C. (2012). WorldMap—a geospatial framework for collaborative research. *Annals of GIS*, 18, 121–134.
- Hecht, B. C., Thrower, F. P., Hale, M. C., Miller, M. R., & Nichols, K. M. (2012). Genetic architecture of migration-related traits in rainbow and steelhead trout, *Oncorhynchus mykiss*. *G3: Genes|Genomes|Genetics*, 2, 1113–1127.
- Henseler, J., & Sarstedt, M. (2013). Goodness-of-fit indices for partial least squares path modeling. *Computational Statistics*, 28, 565–580.
- Huston, M. (1979). A general hypothesis of species diversity. *The American Naturalist*, 113, 81–101.
- IUCN (2015). *The IUCN red list of threatened species*. <https://www.iucnredlist.org>
- Jaswal, A. K., Singh, V., & Bhambak, S. R. (2012). Relationship between sea surface temperature and surface air temperature over Arabian Sea, Bay of Bengal and Indian Ocean. *The Journal of Indian Geophysical Union*, 16, 41–53.
- Lack, D. (1954). *The natural regulation of animal numbers*, London: Oxford University Press.
- Lacy, S. N., Corcoran, D., Alò, D., Lessmann, J., Meza, F., & Marquet, P. A. (2019). Main drivers of freshwater fish diversity across extra-tropical Southern Hemisphere rivers. *Hydrobiologia*, 843, 155–172.
- Legendre, P., & Legendre, L. (2012) *Numerical ecology*. Elsevier.
- Leibold, M. A. (1999). Biodiversity and nutrient enrichment in pond plankton communities. *Evolutionary Ecology Research*, 1, 73–95.
- Lévêque, C., Oberdorff, T., Paugy, D., Stiassny, M. L. J., & Tedesco, P. A. (2008). Global diversity of fish (Pisces) in freshwater. *Hydrobiologia*, 595, 545–567.
- Lieth, H., & Whittaker, R. H. (1975). *Primary productivity of the biosphere*. Springer Verlag.
- Mackenzie, F. T., De Carlo, E. H., & Lerman, A. (2011). Coupled C, N, P, and O biogeochemical cycling at the land-ocean interface. In E. Wolanski & D. S. McLusky (eds), *Treatise on Estuarine and Coastal Science*, vol 5, (pp. 317–342). Waltham: Academic Press.
- Marić, S., Alekseyev, S., Snoj, A., Askeyev, O., Askeyev, I., & Weiss, S. (2014). First mtDNA sequencing of Volga and Ob basin taimen *Hucho taimen*: European populations stem from a late Pleistocene expansion of *H. taimen* out of western Siberia and are not intermediate to *Hucho hucho*. *Journal of Fish Biology*, 85, 530–539.
- Marquet, P. A., Fernandez, M., Navarrete, S. A., & Valdovinos, C. (2004). *Diversity emerging toward a deconstruction of biodiversity patterns. Frontiers of biogeography: New directions in the geography of nature* (pp. 191–209). Sunderland, MA: Sinauer Associates.
- McDowall, R. M. (1988). *Diadromy in fishes*. Timber Press.
- McDowall, R. M. (1997). The evolution of diadromy in fishes (revisited) and its place in phylogenetic analysis. *Reviews in Fish Biology and Fisheries*, 7, 443–462.
- McDowall, R. M. (2002). Accumulating evidence for a dispersal biogeography of southern cool temperate freshwater fishes. *Journal of Biogeography*, 29, 207–219.
- McDowall, R. M. (2008). Why are so many boreal freshwater fishes anadromous? Confronting 'conventional wisdom'. *Fish and Fisheries*, 9, 208–213.
- McLaren, I. (1963). Effects of temperature on growth of zooplankton, and the adaptive value of vertical migration. *Journal of the Fisheries Board of Canada*, 20, 685–727.
- Mims, M. C., Olden, J. D., Shattuck, Z. R., & Poff, N. L. (2010). Life history trait diversity of native freshwater fishes in North America. *Ecology of Freshwater Fish*, 19, 390–400.
- Mittelbach, G. G., Steiner, C. F., Scheiner, S. M., Gross, K. L., Reynolds, H. L., Waide, R. B., Willig, M. R., Dodson, S. I., & Gough, L. (2001). What is the observed relationship between species richness and productivity? *Ecology*, 82, 2381–2396.
- Morais, P., & Daverat, F. (eds) (2016) *An introduction to fish migration*. CRC Press.
- Morales-Barbero, J., & Vega-Álvarez, J. (2019). Input matters matter: Bioclimatic consistency to map more reliable species distribution models. *Methods in Ecology and Evolution*, 10, 212–224.
- Nathan, R., Getz, W. M., Revilla, E., Holyoak, M., Kadmon, R., Saltz, D., & Smouse, P. E. (2008). A movement ecology paradigm for unifying organismal movement research. *Proceedings of the National Academy of Sciences USA*, 105, 19052–19059.
- Nelson, J. S. (2006). *Fishes of the world* (4th ed.). John Wiley & Sons.
- Nyboer, E. A., & Chapman, L. J. (2013). Movement and home range of introduced Nile perch (*Lates niloticus*) in Lake Nabugabo, Uganda: Implications for ecological divergence and fisheries management. *Fisheries Research*, 137, 18–29.
- Oberdorff, T., Guegan, J.-F., Hugué, B., Oberdorff, T., Guegan, J.-F., & Hugué, B. (1995). Global scale patterns of fish species richness in rivers. *Ecography*, 18, 345–352.
- Oberdorff, T., Hugué, B., & Guegan, J.-F. (1997). Is there an influence of historical events on contemporary fish species richness in rivers? Comparisons between Western Europe and North America. *Journal of Biogeography*, 24, 461–467.
- Oksanen, J., Blanchet, G. F., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., Minchin, P. R., O'Hara, R. B., Simpson, G. L., Solymos, P., Henry, M., Stevens, H., Szoecs, E., & Wagner, H. (2017) *vegan: Community ecology package*. <https://CRAN.R-project.org/package=vegan>
- Olson, D., Dinerstein, E., Wikramanayake, E., Burgess, N., Powell, G., Underwood, E., D'Amico, J., Itoua, I., Strand, H., Morrison, J., Loucks, C., Allnutt, T., Ricketts, T., Kura, Y., Lamoreux, J., Wettengel, W., Hedao, P., & Kassem, K. (2001). Terrestrial ecoregions of the world: A new map of life on Earth. *BioScience*, 51, 933–938.
- Oregon State (2017). *Ocean productivity*. <http://sites.science.oregonstate.edu/ocean.productivity/index.php>

- Paradis, E., Claude, J., & Strimmer, K. (2004). APE: Analyses of phylogenetics and evolution in R language. *Bioinformatics*, 20, 289–290.
- Pebesma, E. (2018). Simple features for R: Standardized support for spatial vector data. *The R Journal*, 10, 439–446.
- Pecuchet, L., Lindegren, M., Hidalgo, M., Delgado, M., Esteban, A., Fock, H. O., Gil, L., Payne, M. R., & Punz, A. (2017). From traits to life-history strategies: Deconstructing fish community composition across European seas. *Global Ecology and Biogeography*, 26, 812–822.
- Peiman, K. S., Birnie-Gauvin, K., Midwood, J. D., Larsen, M. H., Wilson, A. D. M., Aarestrup, K., & Cooke, S. J. (2017). If and when: Intrinsic differences and environmental stressors influence migration in brown trout (*Salmo trutta*). *Oecologia*, 184, 375–384.
- Pillar, V. D., & Duarte, L. D. S. (2010). A framework for metacommunity analysis of phylogenetic structure. *Ecology Letters*, 13, 587–596.
- Porch, C. E. (2005). The sustainability of western Atlantic bluefin tuna: A warm blooded fish in a hot blooded fishery. *Bulletin of Marine Science*, 76, 363–384.
- Porto, C. (2015). *World hydrography. Orógenesis soluciones geográficas. Based on shapes from Environmental Systems Research Institute (ESRI)*. <http://tapiquen-sig.jimdo.com>
- Primo, A. L., Azeiteiro, U. M., Marques, S. C., Martinho, F., & Pardal, M. Â. (2009). Changes in zooplankton diversity and distribution pattern under varying precipitation regimes in a southern temperate estuary. *Estuarine, Coastal and Shelf Science*, 82, 341–347.
- Provoost, P., & Bosch, S. (2020). *robis: R client to access data from the OBIS API*. Ocean Biogeographic Information System, Intergovernmental Oceanographic Commission of UNESCO. <https://cran.r-project.org/package=robis>
- Pulido, F. (2007). The genetics and evolution of avian migration. *BioScience*, 57, 165–174.
- Pulido, F. (2011). Evolutionary genetics of partial migration—The threshold model of migration revisited. *Oikos*, 120, 1776–1783.
- Pulido, F., Berthold, P., & van Noordwijk, A. J. (1996). Frequency of migrants and migratory activity are genetically correlated in a bird population: Evolutionary implications. *Proceedings of the National Academy of Sciences USA*, 93, 14642–14647.
- R Core Team (2018). *R: A language and environment for statistical computing*, Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Rabosky, D. L., Chang, J., Title, P. O., Cowman, P. F., Sallan, L., Friedman, M., Kaschner, K., Garilao, C., Near, T. J., Coll, M., & Alfaro, M. E. (2018). An inverse latitudinal gradient in speciation rate for marine fishes. *Nature*, 559, 392–395.
- Radinger, J., & Wolter, C. (2014). Patterns and predictors of fish dispersal in rivers. *Fish and Fisheries*, 15, 456–473.
- Rohde, K. (1992). Latitudinal gradients in species-diversity—The search for the primary cause. *Oikos*, 65, 514–527.
- Ruiz-Jarabo, I., Gonzalez-Wevar, C. A., Oyarzun, R., Fuentes, J., Poulin, E., Bertran, C., & Vargas-Chacoff, L. (2016). Isolation driven divergence in osmoregulation in *Galaxias maculatus* (Jenyns, 1848) (Actinopterygii: Osmeriformes). *PLoS ONE*, 11, 1–21.
- Ruzzante, D. E., Walde, S. J., Gosse, J. C., Cussac, V. E., Habit, E. M., Zemlak, T. S., & Adams, E. D. M. (2008). Climate control on ancestral population dynamics: Insight from Patagonian fish phylogeography. *Molecular Ecology*, 17, 2234–2244.
- Sanchez, G. (2013). *PLS Path Modeling with R*, Berkeley: Trowchez Editions. <http://www.gastonsanchez.com/PLS Path Modeling with R.pdf>
- Schmidt, D. J., Real, K. M., Crook, D. A., & Hughes, J. M. (2013). Microsatellite markers for Australian temperate diadromous fishes *Pseudaphritis urvillii* (Bovichtidae) and *Lovettia sealii* (Galaxiidae). *Conservation Genetics Resources*, 5, 347–349.
- Seibert, H. C. (1949). Differences between migrant and non-migrant birds in food and water intake at various temperatures and photoperiods. *The Auk*, 66, 128–153.
- Shipley, B. (2016). *Cause and correlation in biology* (2nd ed.). : Cambridge University Press.
- Simpson, S. J., Sword, G. A., Lorch, P. D., & Couzin, I. D. (2006). Cannibal crickets on a forced march for protein and salt. *Proceedings of the National Academy of Sciences USA*, 103, 4152–4156.
- Somerville, M., Rodrigues, A. S. L., & Manica, A. (2015). Why do birds migrate? A macroecological perspective. *Global Ecology and Biogeography*, 24, 664–674.
- Southwood, T. (1962). Migration of terrestrial arthropods in relation to habitat. *Biological Reviews*, 37, 171–211.
- Spalding, M. D., Agostini, V. N., Rice, J., & Grant, S. M. (2012). Pelagic provinces of the world: A biogeographic classification of the world's surface pelagic waters. *Ocean and Coastal Management*, 60, 19–30.
- Spalding, M. D., Fox, H. E., Allen, G. R., Davidson, N., Ferdaña, Z. A., Finlayson, M., Halpern, B. S., Jorge, M. A., Lombana, A., Lourie, S. A., Martin, K. D., McManus, E., Molnar, J., Recchia, C. A., & Robertson, J. (2007). Marine ecoregions of the world: A bioregionalization of coastal and shelf areas. *BioScience*, 57, 573–583.
- Svenning, J.-C., Eiserhardt, W. L., Normand, S., Ordonez, A., & Sandel, B. (2015). The influence of paleoclimate on present-day patterns in biodiversity and ecosystems. *Annual Review of Ecology, Evolution, and Systematics*, 46, 551–572.
- Sword, G. A., Lorch, P. D., & Gwynne, D. T. (2005). Insect behaviour: Migratory bands give crickets protection. *Nature*, 433, 703.
- Tinbergen, N. (1963). On aims and methods of ethology. *Zeitschrift für Tierpsychologie*, 20, 410–433.
- Unmack, P. J. (2001). Biogeography of Australian freshwater fishes. *Journal of Biogeography*, 28, 1053–1089.
- Valdovinos, C., Navarrete, S. A., & Marquet, P. A. (2003). Mollusk species diversity in the Southeastern Pacific: Why are there more species towards the pole? *Ecography*, 26, 139–144.
- van der Windt, D. A., & Duarte, L. D. S. (2014). PCPS – An R-package for exploring phylogenetic eigenvectors across metacommunities. *Frontiers of Biogeography*, 6, 144–148.
- van Noordwijk, A. J., Pulido, F., Helm, B., Coppack, T., Delingat, J., Dingle, H. D., Hedenström, A., van der Jeugd, H., Marchetti, C., Nilsson, A., & Pérez-Tris, J. (2006). A framework for the study of genetic variation in migratory behaviour. *Journal of Ornithology*, 147, 221–233.
- Vega, G. C., Pertierra, L. R., & Olalla-Tárraga, M. Á. (2018). Data descriptor: MERRAclim, a high-resolution global dataset of remotely sensed bioclimatic variables for ecological modelling. *Scientific Data*, 5, 180070.
- Vinzi, V. E., Chin, W. W., Henseler, J., & Wang, H. (eds) (2010). *Handbook of partial least squares.*, Springer-Verlag Berlin Heidelberg: Springer Handbooks of Computational Statistics. <https://doi.org/10.1007/978-3-540-32827-81>
- Wegner, N. C., Snodgrass, O. E., Dewar, H., & Hyde, J. R. (2015). Whole-body endothermy in a mesopelagic fish, the opah, *Lampris guttatus*. *Science*, 348, 786–789.
- West, G. C. (1960). Seasonal variation in the energy balance of the tree sparrow in relation to migration. *The Auk*, 77, 306–329.
- Wikelski, M., Tarlow, E. M., Raim, A., Diehl, R. H., Larkin, Ronald, P., & Visser, G. H. (2003). Costs of migration in free-flying songbirds. *Nature*, 423, 704.
- Winemiller, K. O., & Rose, K. A. (1992). Patterns of life-history diversification in North American fishes: Implications for population regulation. *Canadian Journal of Fisheries and Aquatic Sciences*, 49, 2196–2218.
- Woolnough, D. A., Downing, J. A., & Newton, T. J. (2009). Fish movement and habitat use depends on water body size and shape. *Ecology of Freshwater Fish*, 18, 83–91.
- Zhao, M., Heinsch, F. A., Nemani, R. R., & Running, S. W. (2005). Improvements of the MODIS terrestrial gross and net primary production global data set. *Remote Sensing of Environment*, 95, 164–176.
- Zimmerman, J. L. (1965). Bioenergetics of the dickcissel, *Spiza americana*. *Physiological Zoology*, 38, 127–136.
- Zink, R. M. (2011). The evolution of avian migration. *Biological Journal of the Linnean Society*, 104, 237–250.

BIOSKETCH

This paper is one of the chapters from the doctoral thesis of Dominique Alò, which focused on understanding the major ecological and evolutionary processes that drive fish migrations at the local and global level. Dominique's main interests are to understand the mechanisms shaping the distribution of biodiversity, gather evidence and make predictions providing applicable knowledge to the management of vulnerable and endangered species (<https://sites.google.com/view/dominique-alo>).

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Alò D, Lacy SN, Castillo A, Samaniego HA, Marquet PA. The macroecology of fish migration. *Global Ecol Biogeogr.* 2020;00:1–18. <https://doi.org/10.1111/geb.13199>